

Cross-Section Comparison: RS67 vs RS57-70

Analysis Report

May 5, 2026

Live Protons on Target (POT) Summary

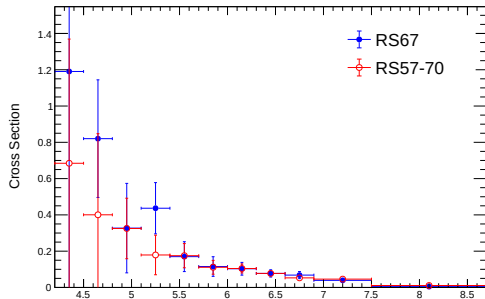
Roadset	LH2 POT	Flask POT	LD2 POT	Total (H2+D2+F)
57	3.53×10^{16}	3.92×10^{15}	1.77×10^{16}	5.69×10^{16}
59	9.37×10^{15}	1.01×10^{15}	4.32×10^{15}	1.47×10^{16}
62	5.65×10^{16}	1.18×10^{16}	2.52×10^{16}	9.35×10^{16}
67	1.61×10^{17}	3.66×10^{16}	7.69×10^{16}	2.75×10^{17}
70	1.79×10^{16}	3.84×10^{15}	8.75×10^{15}	3.05×10^{16}
57–70 Combined	2.80×10^{17}	5.72×10^{16}	1.33×10^{17}	4.70×10^{17}

- POT values extracted from analysis header file.
- RS57–70 used as the primary integrated dataset for comparison.

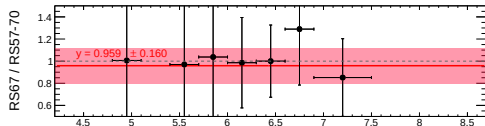
Comparison for $0.0 \leq x_F < 0.05$

Target: LH2

LH2 Cross Section, $0.0 \leq x_F < 0.05$

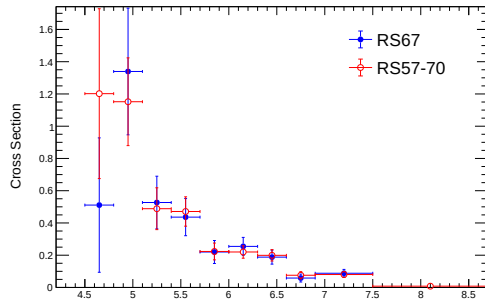


Invariant Mass (GeV)

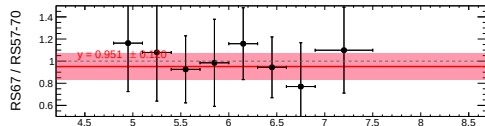


Target: LD2

LD2 Cross Section, $0.0 \leq x_F < 0.05$



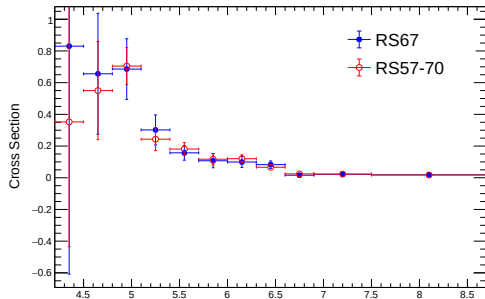
Invariant Mass (GeV)



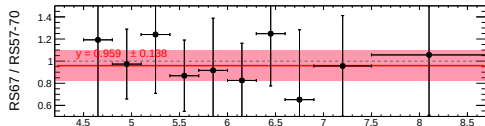
Comparison for $0.05 \leq x_F < 0.1$

Target: LH2

LH2 Cross Section, $0.05 \leq x_F < 0.1$

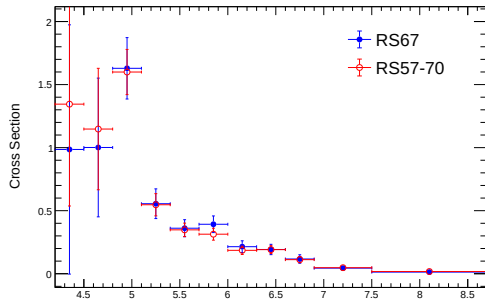


Invariant Mass (GeV)

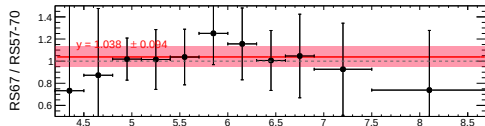


Target: LD2

LD2 Cross Section, $0.05 \leq x_F < 0.1$



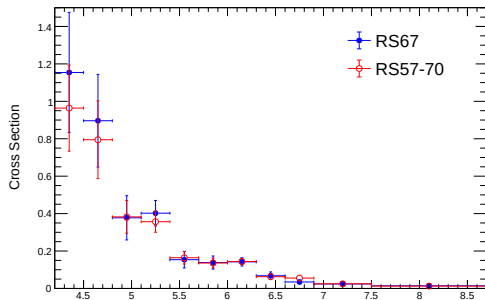
Invariant Mass (GeV)



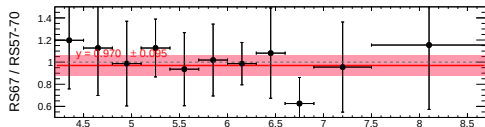
Comparison for $0.1 \leq x_F < 0.15$

Target: LH2

LH2 Cross Section, $0.1 \leq x_F < 0.15$

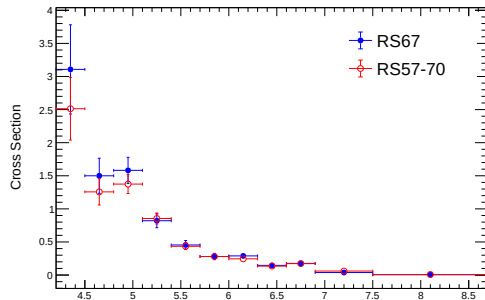


Invariant Mass (GeV)

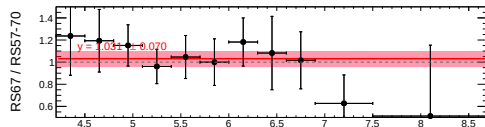


Target: LD2

LD2 Cross Section, $0.1 \leq x_F < 0.15$



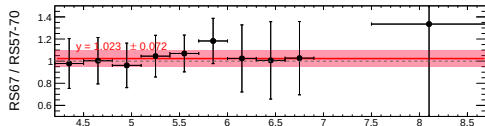
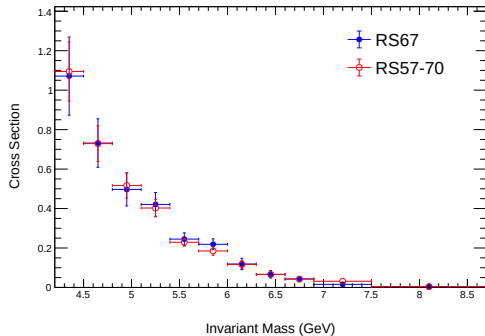
Invariant Mass (GeV)



Comparison for $0.15 \leq x_F < 0.2$

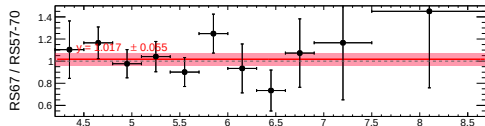
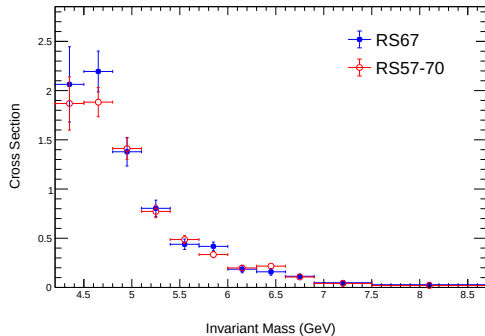
Target: LH2

LH2 Cross Section, $0.15 \leq x_F < 0.2$



Target: LD2

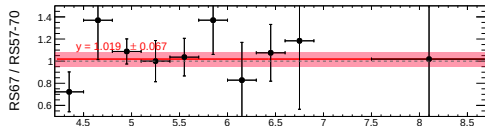
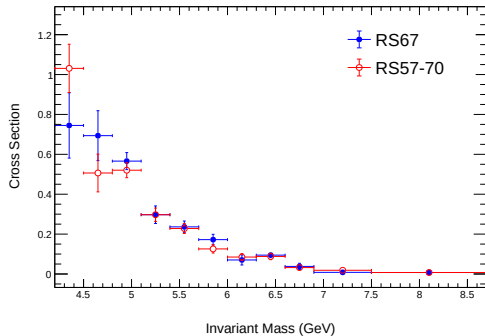
LD2 Cross Section, $0.15 \leq x_F < 0.2$



Comparison for $0.2 \leq x_F < 0.25$

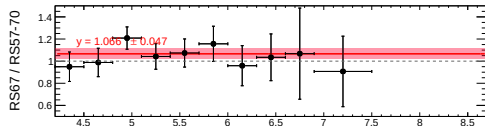
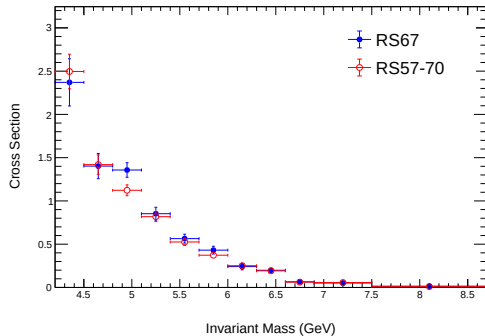
Target: LH2

LH2 Cross Section, $0.2 \leq x_F < 0.25$



Target: LD2

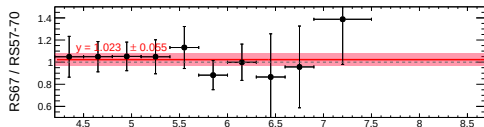
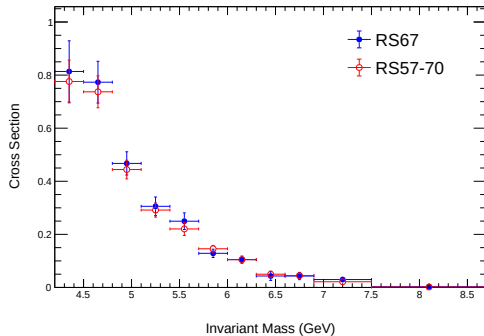
LD2 Cross Section, $0.2 \leq x_F < 0.25$



Comparison for $0.25 \leq x_F < 0.3$

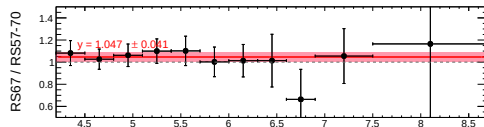
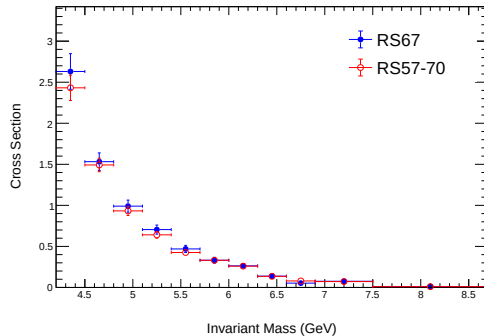
Target: LH2

LH2 Cross Section, $0.25 \leq x_F < 0.3$



Target: LD2

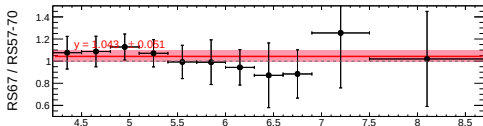
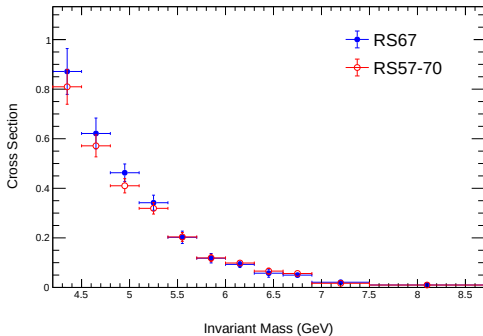
LD2 Cross Section, $0.25 \leq x_F < 0.3$



Comparison for $0.3 \leq x_F < 0.35$

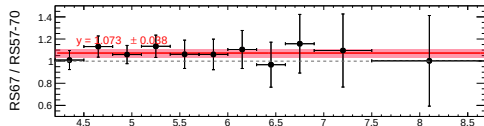
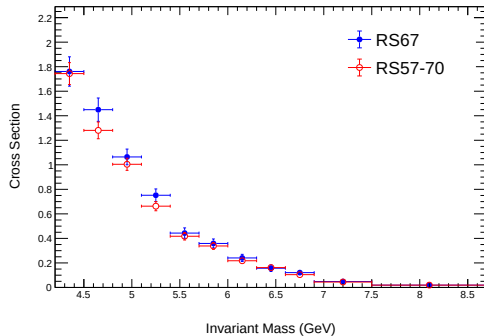
Target: LH2

LH2 Cross Section, $0.3 \leq x_F < 0.35$



Target: LD2

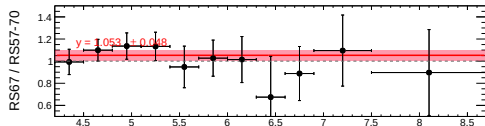
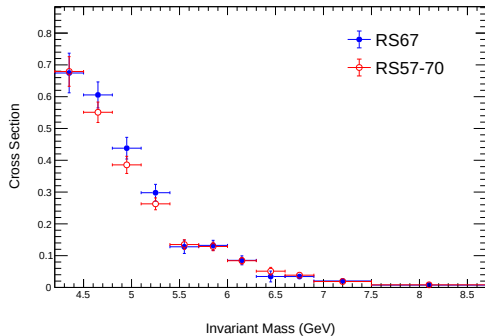
LD2 Cross Section, $0.3 \leq x_F < 0.35$



Comparison for $0.35 \leq x_F < 0.4$

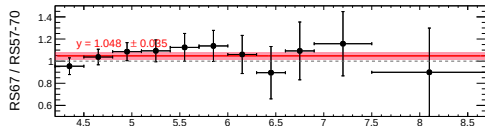
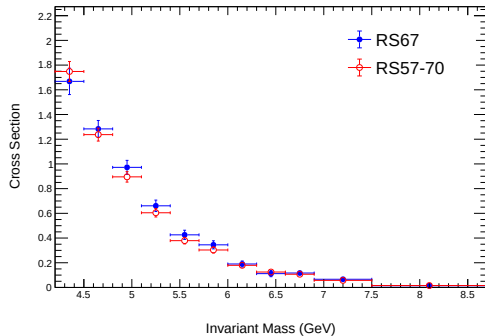
Target: LH2

LH2 Cross Section, $0.35 \leq x_F < 0.4$



Target: LD2

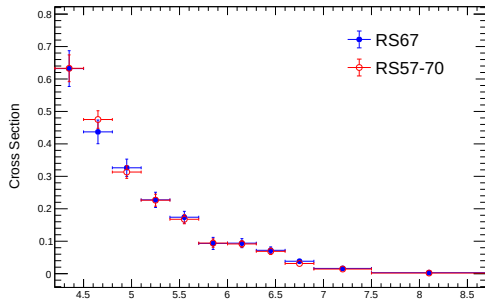
LD2 Cross Section, $0.35 \leq x_F < 0.4$



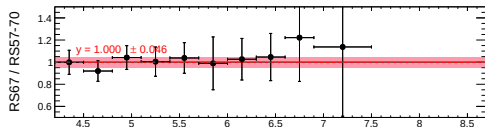
Comparison for $0.4 \leq x_F < 0.45$

Target: LH2

LH2 Cross Section, $0.4 \leq x_F < 0.45$

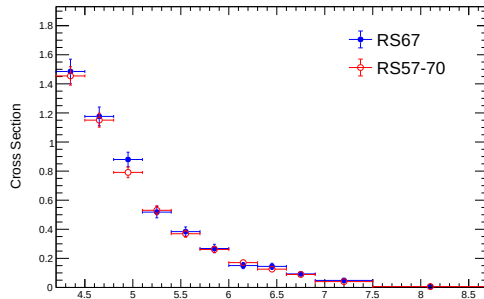


Invariant Mass (GeV)

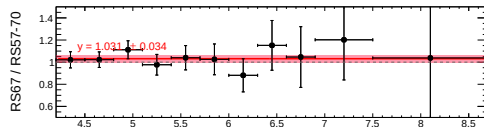


Target: LD2

LD2 Cross Section, $0.4 \leq x_F < 0.45$



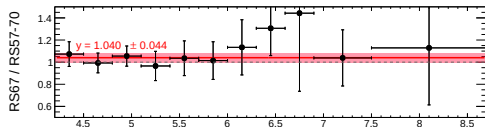
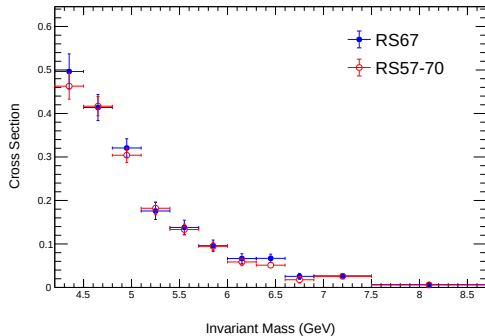
Invariant Mass (GeV)



Comparison for $0.45 \leq x_F < 0.5$

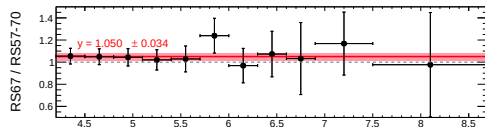
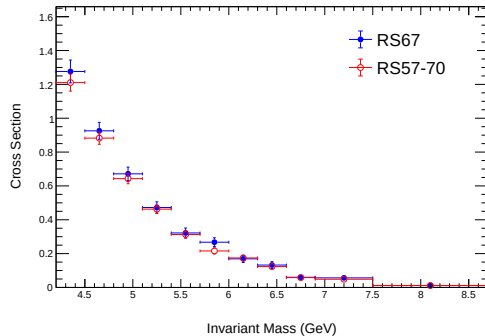
Target: LH2

LH2 Cross Section, $0.45 \leq x_F < 0.5$



Target: LD2

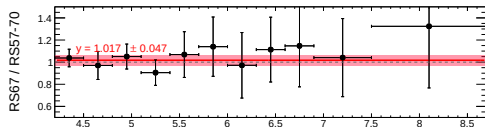
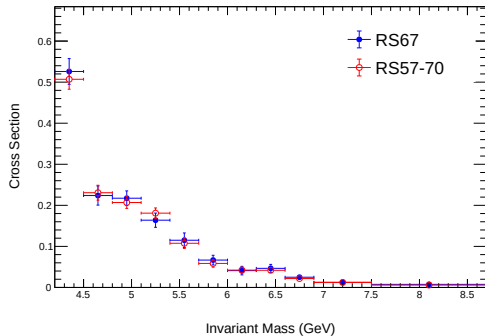
LD2 Cross Section, $0.45 \leq x_F < 0.5$



Comparison for $0.5 \leq x_F < 0.55$

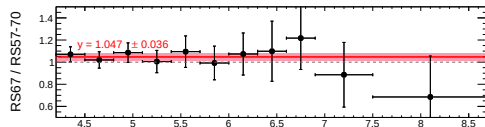
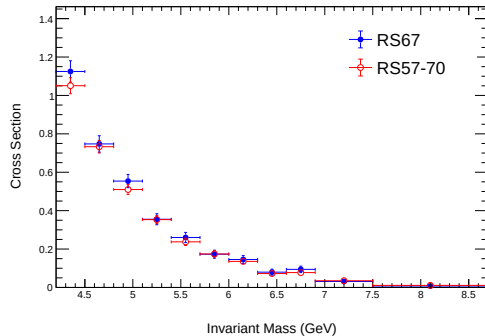
Target: LH2

LH2 Cross Section, $0.5 \leq x_F < 0.55$



Target: LD2

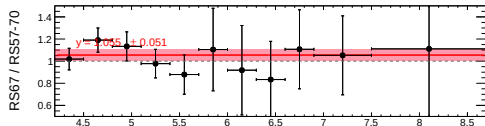
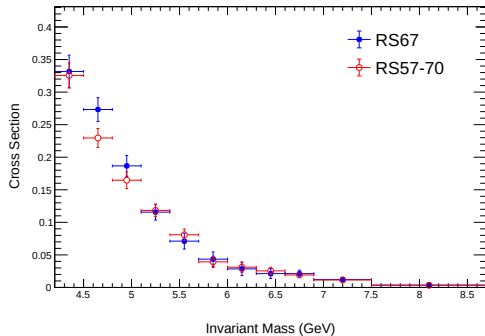
LD2 Cross Section, $0.5 \leq x_F < 0.55$



Comparison for $0.55 \leq x_F < 0.6$

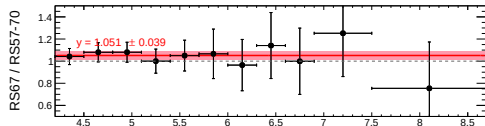
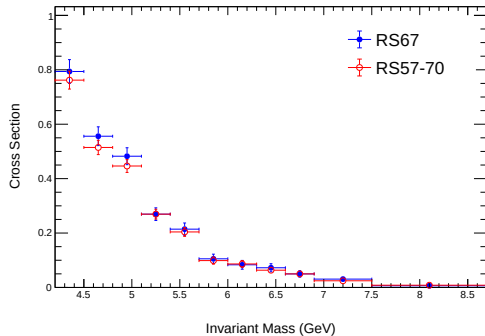
Target: LH2

LH2 Cross Section, $0.55 \leq x_F < 0.6$



Target: LD2

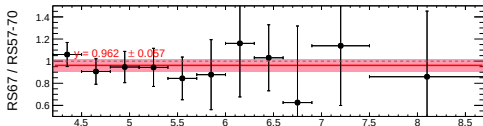
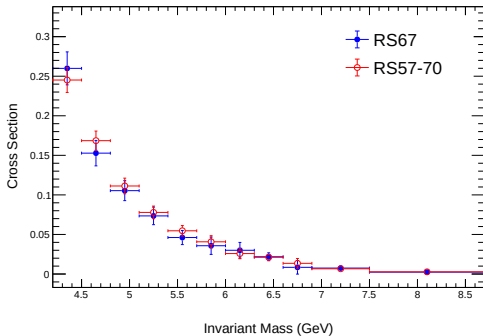
LD2 Cross Section, $0.55 \leq x_F < 0.6$



Comparison for $0.6 \leq x_F < 0.65$

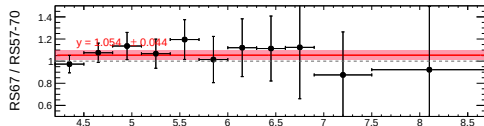
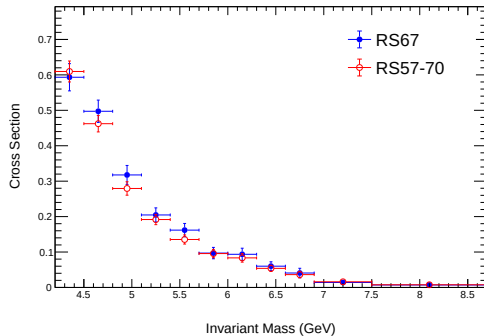
Target: LH2

LH2 Cross Section, $0.6 \leq x_F < 0.65$



Target: LD2

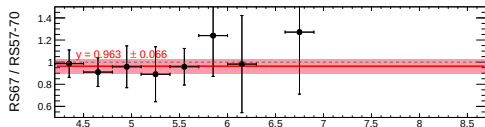
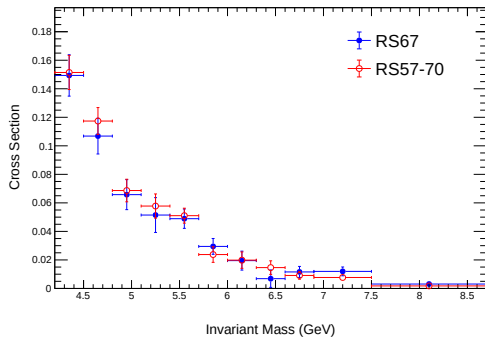
LD2 Cross Section, $0.6 \leq x_F < 0.65$



Comparison for $0.65 \leq x_F < 0.7$

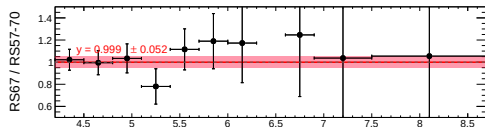
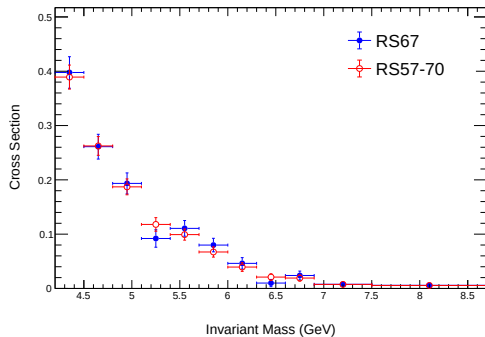
Target: LH2

LH2 Cross Section, $0.65 \leq x_F < 0.7$



Target: LD2

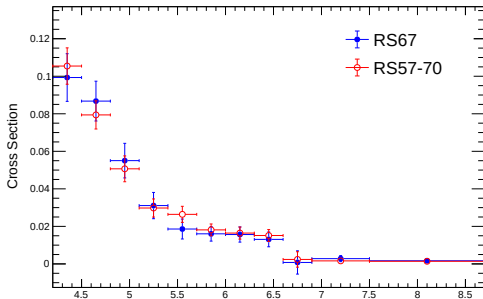
LD2 Cross Section, $0.65 \leq x_F < 0.7$



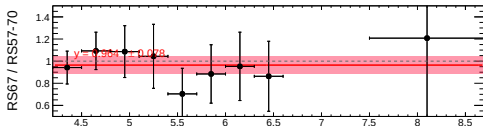
Comparison for $0.7 \leq x_F < 0.75$

Target: LH2

LH2 Cross Section, $0.7 \leq x_F < 0.75$

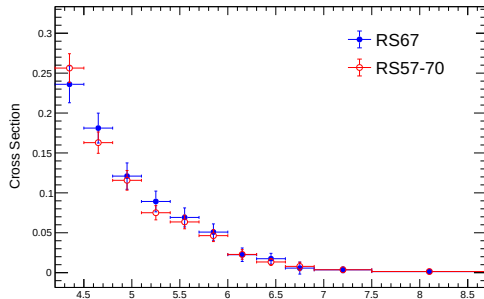


Invariant Mass (GeV)

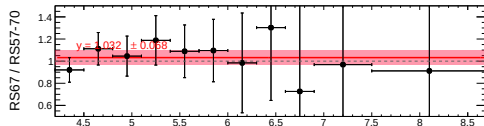


Target: LD2

LD2 Cross Section, $0.7 \leq x_F < 0.75$



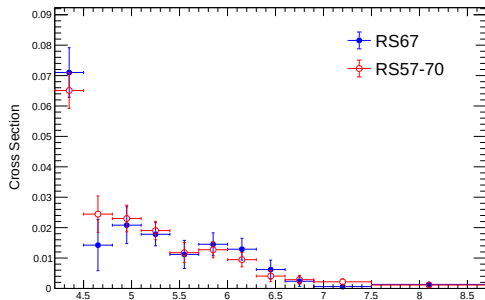
Invariant Mass (GeV)



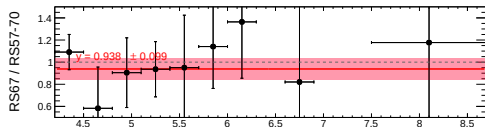
Comparison for $0.75 \leq x_F < 0.8$

Target: LH2

LH2 Cross Section, $0.75 \leq x_F < 0.8$

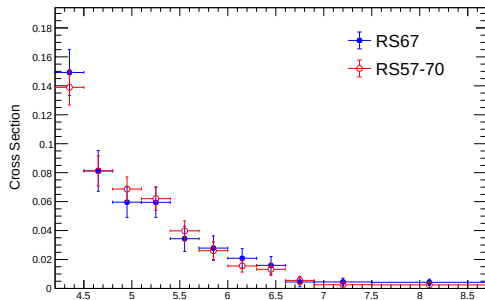


Invariant Mass (GeV)



Target: LD2

LD2 Cross Section, $0.75 \leq x_F < 0.8$



Invariant Mass (GeV)

